

Satwik Bhattamishra

PhD student, University of Oxford

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Education

Present Oct 2021	University of Oxford Ph.D. in Computer Science • Google DeepMind Scholarship Advisors: Prof. Phil Blunsom and Prof. Varun Kanade	Oxford, UK
May 2019 Aug 2014	Birla Institute of Technology and Science Pilani B.E. (Hons.), Computer Science and Int. M.Sc. (Hons.), Biological Science	Pilani, India

Industry Experience

Mar 2025 Dec 2024	Google <i>Student Researcher / Cloud AI Research Team</i> Explored scaling test-time compute and training for LLM agents. Explored various decoding and self-verification strategies to improve per-step action prediction accuracy of agents. Tools: PyTorch, GPUs	Sunnyvale, US
Oct 2024 June 2024	Cohere <i>Intern of Technical Staff / Foundations team</i> Pretrained LLMs with <i>hybrid</i> architectures involving SSMS and Transformers. Pretrained 7B models, optimized inference strategies, & integrated μ P for hyperparameter transfer. Tools: JAX, TPUs	London, UK
Oct 2023 June 2023	<i>Intern of Technical Staff / Foundations team</i> Pretrained and evaluated LLMs with state-space and long-convolutional architectures. Pretrained 7B models and conducted ablations to evaluate the efficacy of architectural components. Tools: JAX, TPUs	
July 2021 July 2019	Microsoft Research <i>Research Fellow / Advisors: Dr. Navin Goyal, Dr. Monojit Choudhury</i> Analyzed the computational capabilities of Transformers and LSTMs to model various classes of formal languages; worked on compositional generalization and math word problem solving. Tools: PyTorch, GPUs	Bangalore, India

Publications

Provably Learning Attention with Queries Satwik Bhattamishra , Kulin Shah, Michael Hahn, Varun Kanade 2026 International Conference on Machine Learning	[paper] [ICML '26]
Discovering Interpretable Algorithms by Decompiling Transformers to RASP Xinting Huang, Aleksandra Bakalova, Satwik Bhattamishra , William Merrill, Michael Hahn 2026 International Conference on Machine Learning	[paper] [ICML '26]
Automata Learning and Identification of the Support of Language Models Satwik Bhattamishra , Michael Hahn, Varun Kanade 2026 International Conference on Learning Representations	[paper] [ICLR '26]
Benefits and Limitations of Communication in Multi-Agent Reasoning Michael Rizvi-Martel, Satwik Bhattamishra , Neil Rathi, Guillaume Rabusseau, Michael Hahn 2026 International Conference on Learning Representations	[paper] [ICLR '26]
The Transformer Cookbook A. Yang, C. Watson, A. Xue, Satwik Bhattamishra , J. Llerena, W. Merrill, E. Ferreira, A. Svete, D. Chiang 2026 Transactions on Machine Learning Research	[paper] [TMLR '26]
A Formal Framework for Understanding Length Generalization in Transformers X. Huang, A. Yang, Satwik Bhattamishra , Y. Sarrof, A. Krebs, H. Zhou, P. Nakkiran, M. Hahn 2025 International Conference on Learning Representations	[paper] [ICLR '25]
Separations in the Representational Capabilities of Transformers and Recurrent Architectures Satwik Bhattamishra , Michael Hahn, Phil Blunsom, Varun Kanade 2024 Conference on Neural Information Processing Systems	[paper] [NeurIPS '24]
Understanding In-Context Learning in Transformers by Learning to Learn Discrete Functions Satwik Bhattamishra , Arkil Patel, Phil Blunsom, Varun Kanade 2024 International Conference on Learning Representations [Oral, Top 1.2%]	[paper] [ICLR '24]

Inshrinkerator: Compressing Deep Learning Training Checkpoints via Dynamic Quantization A. Agrawal, S. Reddy, Satwik Bhattamishra , V. Sarath Nookala, V. Vashishth, K. Rong, A. Tumanov 2024 ACM Symposium on Cloud Computing	[paper] [SoCC '24]
Simplicity Bias in Transformers and their Ability to Learn Sparse Boolean Functions Satwik Bhattamishra , Arkil Patel, Varun Kanade, Phil Blunsom 2023 Annual Meeting of the Association for Computational Linguistics	[paper] [ACL '23]
MAGNIFICO: Evaluating the In-Context Learning Ability of Large Language Models to Generalize to Novel Interpretations Arkil Patel, Satwik Bhattamishra , Siva Reddy, Dzmitry Bahdanau 2023 Conference on Empirical Methods in Natural Language Processing [Oral]	[paper] [EMNLP '23]
Revisiting the Compositional Generalization Abilities of Neural Sequence Models Arkil Patel, Satwik Bhattamishra , Phil Blunsom, Navin Goyal 2022 Annual Meeting of the Association for Computational Linguistics	[paper] [ACL '22]
Are NLP Models really able to Solve Simple Math Word Problems? Arkil Patel, Satwik Bhattamishra , Navin Goyal 2021 Conference of the North American Chapter of the Association for Computational Linguistics	[paper] [NAACL '21]
On the Ability and Limitations of Transformers to Recognize Formal Languages Satwik Bhattamishra , Kabir Ahuja, Navin Goyal 2020 Conference on Empirical Methods in Natural Language Processing	[paper] [EMNLP '20]
On the Practical Ability of Recurrent Neural Networks to Recognize Hierarchical Languages Satwik Bhattamishra , Kabir Ahuja, Navin Goyal 2020 International Conference on Computational Linguistics [Best Short Paper Award]	[paper] [COLING '20]
On the Computational Power of Transformers and Its Implication in Sequence Modeling Satwik Bhattamishra , Arkil Patel, Navin Goyal 2020 Conference on Computational Natural Language Learning	[paper] [CoNLL '20]
Submodular Optimization-based Diverse Paraphrasing and its Effectiveness in Data Augmentation Ashutosh Kumar*, Satwik Bhattamishra *, Manik Bhandari, Partha Talukdar (* = Equal Contribution) 2019 Conference of the North American Chapter of the Association for Computational Linguistics [Oral]	[paper] [NAACL '19]

Selected Dev Projects

Student Developer Google Summer of Code (GSoC) 2016	May'16 - Aug'16
Worked on the Phenopacket scraper project, which extracts information from texts scraped from life sciences websites, analyzes them, and generates a phenopacket based on the correct ontology references. [Project Page]	
Machine Learning Contests [Kaggle Profile]	
Kaggle Level: Competitions Expert. Silver medal in Kaggle Santander Value Prediction Challenge, Rank: Pvt. 185 th Pub. 189 th /4484. Bronze medal in Kaggle Instacart Competition, Rank: Pvt. 195 th Pub. 74 th /2623. Qualified for Zonal Round in India Hacks Machine Learning Competition by HackerEarth, Rank: 29 th /860.	
Review Miner Microsoft Code.Fun.Do Hackathon [GitHub]	
Developed a cross-platform application which analyses reviews from commercial websites and provides insights about products based on keyword extraction and sentiment analysis. Winner of Hackathon at BITS Pilani, Rank 1 st /90+ teams.	

Selected Talks

Representational Capabilities of Transformers and Recurrent Architectures	[Video]
Charles University, Czech Republic [10/25] Saarland University, Germany [07/25]	
Language Modelling with Recurrent and State Space Architectures	[Video]
Georgia Tech (SysML Guest Lecture) [11/24]	
On the Ability of Neural Sequence Models to Recognize Formal Languages	[Slides]
DeepMind NLP Reading Group [03/22] FLANN [02/22] MALL Lab, IISc [12/20]	

Teaching and Services

Teaching Assistant ×2 Computational Learning Theory, Oxford	Fall 22 and Fall 23
Teaching Assistant Neural Networks and Fuzzy Logic, BITS Pilani	Jan'18 - May'18
Reviewer ICML 2024, 2023; NeurIPS 2023, 2022; ACL 2023, 2022; ICLR 2022; EMNLP 2022, 2021, 2020; NAACL 2021	